

Stimulus- and Response-locked Posterior Contralateral Negativity Bisect Cognitive Operations in Visual Search

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Abstract

■ We explored the flow of information during visual search by examining activity indexing visual attention (N2pc) and the subsequent processing of the selected objects in visual short-term memory (SPCN) time-locked to stimulus presentation and to the motor response. We measured event-related activity at posterior sites (PO7/PO8) for 96 participants during a simple visual search task. A response-locked posterior contralateral negativity (RLpcN) was observed with a scalp distribution similar to that of the N2pc and SPCN. The RLpcN was compared with the stimulus-locked activity (N2pc and SPCN) across experimental manipulations (targets were either closer or farther from fixation in visual space, and the response was either more frequent [75%] or less frequent [25%]) and across response speeds (EEG data were separated into tertiles

by RT both within-subjects and between-subjects). The leading edge and early portion of the RLpcN appeared to reflect the initial deployment of attention (N2pc), whereas the later portion (up to peak amplitude) reflected subsequent processing of visual information (SPCN). SPCN and RLpcN also had similar modulations in amplitude for both analyses. Moreover, whereas very small N2pc and SPCN onset latency differences were observed when data were separated into tertiles by RT, there were large onset differences for the RLpcN, with earlier RLpcN onsets for longer RTs, suggesting that RT variance is in large determined by processing after the initial deployment of attention. The results show how we can bisect processing responsible for variations in RT relative to the onset of visual spatial attention. ■

INTRODUCTION

When using event-related potentials (ERPs) to study the visual system, researchers typically observe and assess sensory, perceptual, and cognitive processing by segmenting and averaging relative to the onset of a visual display. In particular, the moment of visual stimulus presentation is of paramount importance to study early visual components (e.g., the P1, N1, and N2; Berchicci, Spinelli, & Di Russo, 2016; Hillyard & Kutas, 1983), and their rapid succession after stimulus onset is thus observed with little variance in component onset between trials. However, more downstream (later in time) components, generally reflecting later and more cognitive stages of processing, are also influenced by endogenous factors (e.g., task instruction or difficulty in comparison with exogenous factors such as stimulation), and their average is smeared by the trial-by-trial variance accrued since stimulus presentation (Luck, 2014). Later components could also be associated to events other than the presentation of stimulation or stimulus evaluation, such as response

selection (lateralized readiness potential [LRP]; see Eimer, 1998). Earlier work has examined a variety of ERP components (e.g., the P3, N2, and P2; Berchicci et al., 2016; LRP: Osman, Moore, & Ulrich, 1995) time-locked to both the stimulus and motor response to track the dynamics of processing (see also Verleger, Jaśkowski, & Wascher, 2005). Components time-locked to both the response and stimulus have also been used to partition the stream of cognitive processing into operations occurring before and after a particular operation. For example, stimulus- and response-locked LRPs (S-LRP and R-LRP, respectively) have been used to examine processing taking place during response selection (or before) separately from those taking place after response selection (i.e., response execution). Osman et al. (1995) used this approach to show that precuing could affect the interval between R-LRP onset and the response, suggesting a speeding up of response execution.

Researchers have also examined components indexing different processes and alongside behavioral measures together to understand better the flow of information from stimulus to response during visual search (e.g., Cosman, Arita, Ianni, & Woodman, 2016; Töllner, Rangelov, & Müller, 2012; Hackley, Schankin, Wohlschlaeger, & Wascher, 2007). Indeed, Hackley et al. (2007) demonstrated

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that, when more time was given to interpret a cue (600 vs. 3000 msec before stimulus presentation), RTs and the time between N2pc onset and S-LRP onset were both shorter, whereas no significant latency effects were observed between stimulus presentation and N2pc onset latency or between the onset of the R-LRP to the response. Moreover, Töllner et al. (2012) examined activity from four different search task conditions all having physically identical stimuli (target localization, detection, discrimination, or compound responses¹). Similar to Hackley et al. (2007), Töllner et al. (2012) observed no differences between tasks in N2pc onset latency, whereas the onset of the S-LRP mirrored behavioral RTs: Localization tasks were performed with the fastest RTs and produced the earliest S-LRP, followed by discrimination, detection, and compound response tasks, in that order. Thus, the results of Hackley et al. (2007) and Töllner et al. (2012) suggest that variations in RTs, in their particular tasks, reflected principally operations taking place after the onset of the N2pc and before the onset of the S-LRP (e.g., stimulus categorization and response selection).

In the present work, we examined lateralized posterior neural activity during visual search associated with visual spatial attention, indexed by the N2pc (Luck & Hillyard, 1994b), and the processing of selected visual objects in visual short-term memory (VSTM), indexed by the SPCN component (Jolicœur, Brisson, & Robitaille, 2008; Vogel & Machizawa, 2004), alongside mental chronometry (RTs). We compared the results of the usual stimulus-locked averaging method for N2pc and SPCN with their response-locked counterpart. We wished to demonstrate that combining stimulus- and response-locked analyses could provide a way to bisect the sequence of operations into those that influence processing before the engagement of attention versus those that influence processing after. Thus, we provide converging evidence with the aforementioned work considering relationships among multiple components (Cosman et al., 2016; Töllner et al., 2012; Hackley et al., 2007) and attribute lateralized activity associated with processing occurring before and after the deployment of attention.

The N2pc is a posterior, lateralized, ERP component characterized by a greater negativity contralateral to a potential target relative to the ipsilateral response and is typically observed approximately 200–300 msec after stimulus onset (Luck & Hillyard, 1994b). Whether this component reflects the selection of a target (Eimer, 1996) or the suppression of distractors (Luck & Hillyard, 1994a, 1994b) is still debated. The N2pc has been isolated by time-locking ERP averages relative to the onset of visual displays in which the typical stimulus of interest (i.e., the target) is a pop-out stimulus (very easy to locate) in the left or right visual field. In this preparation, the N2pc is strongly synchronized to stimulus onset with little variance in component onset across trials. As such, differences in N2pc (i.e., latency, amplitude, and/or scalp distribution) are attributed to task manipulations (Luck & Kappenman, 2012). We also examined the SPCN² (Vogel & Machizawa,

2004), a contralateral negativity often beginning roughly 300–400 msec after stimulus onset at posterior electrodes (similar to the N2pc). The SPCN is dissociable from the N2pc (Jolicœur et al., 2008) and is suggested to reflect the maintenance of information and/or subsequent processing of visual objects in VSTM (Jolicœur et al., 2008; Dell'Acqua, Sessa, Jolicœur, & Robitaille, 2006; Vogel & Machizawa, 2004). The goal of the present work was to identify processing stages associated with variations in RT using a comparison of stimulus-locked indices of visual attention (N2pc) and passage into VSTM (SPCN) with analyses of response-locked visual-spatial posterior contralateral negativity (RLpcN). We wished to observe directly in measured ERPs the influence of variations in processing time on processing that follows the engagement of visual spatial attention.

Examining ERPs calculated relative to different time-locking events can offer some general advantages. Averaging techniques can suffer from a so-called magnifying glass effect: An increase in resolution of specific ERPs can sometimes be at the cost of others. For example, in a stimulus-locked average, activity not time-locked to stimulus onset could be distorted or disappear in the average (Luck, 2014). Although several alternative methods have been proposed, for example, researchers will try and improve the resolution and validity of their results by performing more complex statistical analyses (i.e., independent component analysis [ICA], principal component analysis, linear modeling, residue iteration decomposition; for the latter, see Ouyang, Sommer, & Zhou, 2015; Ouyang, Herzmann, Zhou, & Sommer, 2011), these methods generally make several strong assumptions regarding the data and are not always suitable (see Poli, Cinel, Citi, & Sepulveda, 2010). We therefore suggest that comparing results from various angles, such as from the stimulus and response, is an alternative and effective way to capture better the underlying activity of interest and that this approach has the potential to correct, in part, the magnifying glass effect.

We suggest that this exploratory approach will provide information regarding where processing associated with variations in RTs occurs. The task was a simple visual search task with a pop-out target: Participants located an orange or green target among blue distractors (all stimuli were squares with a gap on one side) and indicated whether the gap was on the top or not. Thus, we suppose that target processing included at least four phases: (a) locating the pop-out stimulus on the basis of color and deploying attention to this item, (b) locating the key task-relevant attribute (i.e., the location of the gap), (c) classifying this location and selecting the appropriate response based on task instructions, and (d) response execution (leading to the button press). Two separate analyses were conducted: The first compared waveforms observed in different experimental conditions, and the second separated ERPs into tertiles by RTs on a between-subjects basis (fast participants, middle

participants, and slow participants) and a within-subjects basis (fast trials, middle trials, and slow trials). In the response-locked average, we expected a larger negativity contralateral to the target side (relative to the ipsilateral response) with a scalp distribution similar to the N2pc and SPCN. We expected that the onset of the response-locked contralateral negativity (RLpcN) would provide information concerning processes taking place during and after the initial engagement of attention on the pop-out stimulus. An increase in the duration of processing after the deployment of attention should result in an earlier RLpcN onset relative to the response, in other words, a longer interval between the leading edge of the RLpcN and response execution. We also expected that RLpcN peak amplitude would reflect response-locked SPCN activity, because the SPCN occurs later in the processing pipeline than the N2pc and is therefore temporally closer to the response. In particular, we expected that an increase in need for subsequent processing of selected visual objects in VSTM would result in larger RLpcN and SPCN amplitudes.

The first analysis examined the data with a 2×2 experimental design: (a) The target was either near or far from fixation, and (b) the response was either less frequent (gap in the square is on the top side, $p = .25$) or more frequent (gap in the square is on one of the other three sides, $p = .75$). With regard to the distance from fixation, Schaffer, Schubö, and Meinecke (2011) found that N2pc was sensitive to target eccentricity. In their experiment, target stimuli were presented with a retinal eccentricity of 0° , 1.7° , 3.5° , 5.2° , and 7° along the horizontal meridian. N2pc amplitude increased as distance to fixation decreased (from 7° to 1.7°). Moreover, using the same paradigm and a portion of the participants (22 participants) as the present work, previous work from our laboratory also found a larger N2pc for targets nearer fixation compared with targets farther from fixation (West et al., 2015), and so we expected the same result. To our knowledge, no work has examined the effect of target eccentricity for the SPCN. We hypothesized that an increase in activity associated with perceptual attention (N2pc) for closer targets would likely lead to greater activity associated with the subsequent processing in VSTM (SPCN). Moreover, because we hypothesized that the RLpcN amplitude represents response-locked SPCN activity, the same effect should also be observed for the RLpcN. For the frequency of response manipulation, we had no a priori reasons to expect differences in N2pc, because this manipulation was expected to influence mainly processing after the engagement of attention on the target. However, we expected longer RTs for less frequent responses compared with more frequent responses, reflecting a difference in time required to, for example, select a response after the engagement of spatial attention on the target. Thus, this effect should be visible in the onset latency of the RLpcN, with an earlier onset for less frequent compared with more frequent

responses, because more time should elapse between the onset of attentional engagement and the response. We also expected larger SPCN and RLpcN amplitudes for less frequent compared with more frequent responses. This expectation hinged on the hypothesis that more processing would be required for the less frequent response (the more novel condition) after the initial deployment of attention to alter preparation for the more frequent response. This adjustment should require additional neural activity in subsequent processing, such as greater dependence on VSTM (e.g., Maheux & Jolicoeur, 2017).

The second analysis sorted epochs into tertiles by RT on a trial-by-trial basis for each participant (and within each condition; within-subjects analysis) as well as on a between-subjects basis based on their overall mean RT. In a study using a large subset of the current dataset (83 of 96 participants), Drisdelle, West, and Jolicoeur (2016) examined stimulus-locked ERPs split into fast and slow participants as well as fast and slow trials (for every participant) based on the median RT. A larger N2pc was observed for fast participants and trials, possibly suggesting a relationship between N2pc amplitude and the eventual RT. Recent research has also found a larger N2pc component with practice (also marked by a decrease in RTs; Clark, Appelbaum, van den Berg, Mitroff, & Woldorff, 2015). Although an earlier N2pc component for shorter RTs was also observed, the difference in behavioral RTs between the fast and slow participants or trials (Drisdelle et al., 2016) and between the first and last sessions (before and after practice; Clark et al., 2015) was much larger than the corresponding N2pc latency onset difference, suggesting that most of the processing speed improvement likely occurred downstream (see also Wolber & Wascher, 2005). Given that we are using a large portion of the data analyzed in Drisdelle et al. (2016), we expected that N2pc amplitude will be larger and onset latency will be slightly earlier for shorter RTs. We therefore expected that activity reflecting most of the RT variance would occur mainly during processing downstream of N2pc onset. Where the current work goes beyond Drisdelle et al. (2016) is in the examination of the RLpcN. We expected the leading edge (onset) of the RLpcN would start earlier relative to the response for longer RTs. This pattern would reflect processing taking place after the initial engagement of attention, leading up to the response. For SPCN and RLpcN amplitudes, we had no strong a priori hypothesis for RT tertile split effects. We considered two possible outcomes: (1) no difference in amplitude, indicating that the information entering and sustained in VSTM does not vary with RT, and (2) an increase in amplitude with decreasing RT, similar to the N2pc, suggesting that a larger activity associated with visual attention produced a more detailed representation in VSTM or that, when target identification was more difficult (for any number of reasons; reflected by longer RTs), they are carried forward (i.e., beyond the initial perceptual processing, reflected by the N2pc) for further processing.

Regardless of outcome, and importantly, we expected that both SPCN and RLpcN behave in the same manner, demonstrating that RLpcN amplitude is response-locked SPCN activity.

METHODS

Participants

One hundred fifty-four (154) participants were compensated CAD 20 for voluntarily participating in the experiment, which was vetted by the ethics committee of the Faculty of Arts and Science at Université de Montréal. Data from some participants used in the present work were also included in previous work (22 participants from West et al., 2015; 83 participants from Drisdelle et al., 2016, some of which overlap). Only data from participants who reported being nonaction video game players from the West et al. (2015) study were included in the present analyses. All participants kept for final analysis, a total of 96 participants (age: $M = 23.4$ years, $SD = 3.4$ years; 61 women; all right-handed), reported no neurological or psychiatric problems, were not taking psychoactive medication, and reported having normal or corrected-to-normal vision and normal color vision. Before beginning the task, participants filled out a questionnaire regarding personal information and provided written informed consent.

Procedure

Participants were seated in a dimly lit room at a viewing distance of 57 cm from a computer screen. They completed the visual search task described below. The experimental procedure and stimuli were programmed and run using E-Prime (E-Prime 1.2, n.d. [Version 1.2.1.844]).

Visual Stimuli

As illustrated in Figure 1, the visual search display consisted of colored squares (size: $1.5^\circ \times 1.5^\circ$) each containing a gap on one side (gap size: 0.5°). The gap could be presented on the top, left, right, or bottom side of the square with equal probability ($p = .25$). In each visual display, two squares were presented in the left visual field and two were presented in the right visual field. The target could be in any of these four positions with equal probability ($p = .25$). A plus (+) sign at the center of the screen served as a fixation cross. The center of the squares closer to fixation were 2.25° below and 3.00° lateral (left or right) from fixation. The center of the squares farther from fixation were 3.75° below and 5.25° lateral (left or right) from fixation. The target was defined by color. In target discrimination trials, there were two target colors whose probability of presentation was manipulated: The target was either orange (more frequent color: $p = .80$) or green (less frequent color: $p = .20$).

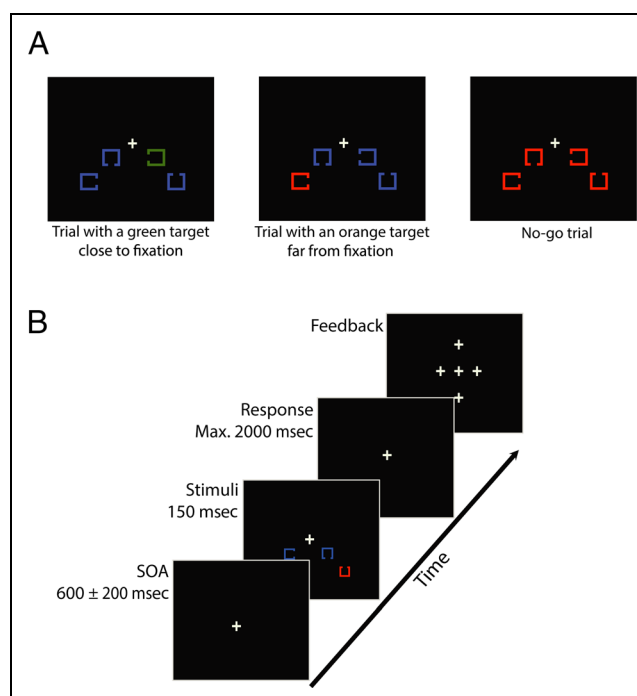


Figure 1. Example stimuli and time course in a trial. The task was to respond based on the gap location in the unique colored square (participant indicated whether the gap was on the top side or not) by pressing a key or to make no response if all the squares were of the same color. (A) Examples of stimulus displays (green near target on the left, orange far target in the middle, and no-go trials on the right). (B) Illustration of the time course of a trial.

The three remaining squares (distractors) were blue. To control for low-level sensory responses, the colors were adjusted to have a similar luminance using a chromometer (Minolta CS100, Konica Minolta; orange [9 cd/m^2 ($0.545 \text{ } 0.384$)], green [9.1 cd/m^2 ($0.311 \text{ } 0.569$)], and blue [8.4 cd/m^2 ($0.159 \text{ } 0.113$)]). The relative frequency of the target color had no impact on the results (no effects on N2pc, SPCN, and RLpcN) and thus was not useful in the present investigation so the data were collapsed. The two manipulations of interest were response frequency and distance of the target from fixation.

Visual Search Task

Two types of trials were presented: target discrimination trials (83.3% of experimental trials) and no-go trials (16.7% of experimental trials). For no-go trials, all four squares were orange and participants were instructed not to respond (see Figure 1A). Because we were interested in response-locked averages, no-go trials were not analyzed.

Target Discrimination Task

When a target was present, participants indicated the location of the gap (gap on the top side of the square

or not). The gap was presented with equal probability on each side of the target square ($p = .25$). Therefore, the response was either more frequent (gap on the bottom side, left side, or right side, $p = .75$) or less frequent (gap on the top side, $p = .25$). Participants responded by pressing either the “C” key with the left index finger or the “M” key with the right index finger for either more frequent or less frequent response trials (keys were counterbalanced between participants). A fixation cross was presented at the center of the screen for the entire experiment, and participants were instructed to maintain fixation and refrain from blinking during trials. Trials were initiated when the participant pressed the spacebar. Following an average of 600 msec (± 200 msec jitter) after trial initiation, the visual search display was presented for 150 msec. After the presentation of the visual search display, participants had 2000 msec to respond. Once a response was executed, feedback was given: Participants were presented with a cross made of five plus symbols (+) for correct responses, five minus symbols (–) for incorrect responses, or five vertical bars (|) if they did not respond in the allotted 2000 msec (see Figure 1B). Feedback was presented until the participant initiated the following trial. An example of a target discrimination trial sequence is illustrated in Figure 1B. Participants first completed a practice block (40 trials) and then eight experiment blocks (96 trials each; total experimental trials: 768).

Data Analysis and Statistical Analyses

EEG Recordings

The EEG was recorded at a sampling rate of 512 Hz (online low-pass antialiasing filter: 104 Hz) with 64 Ag–AgCl electrodes using the International 10–10 system (at the following sites: Fp1, Fpz, Fp2, AF7, AF3, AFz, AF4, AF8, F7, F5, F3, F1, Fz, F2, F4, F6, F8, FT7, FC5, FC3, FC1, FCz, FC2, FC4, FC6, FT8, T7, C5, C3, C1, Cz, C2, C4, C6, T8, TP7, CP5, CP3, CP1, CPz, CP2, CP4, CP6, TP8, P9, P7, P5, P3, P1, Pz, P2, P4, P6, P8, P10, PO7, PO3, POz, PO4, PO8, O1, Oz, O2, and Iz; Sharbrough et al., 1991) placed on an elastic cap. Data were recorded using a Biosemi ActiveTwo system and ActiView acquisition software (BioSemi B. V.). Two external electrodes placed on the left and right mastoids were averaged and used for offline referencing. To measure horizontal eye movements (such as saccades), a horizontal EOG (HEOG) was recorded, and the difference in voltage between two external electrodes placed lateral to the external canthi was calculated. To measure eye blinks and vertical eye movements, a vertical EOG was recorded, and the difference in voltage between external electrodes placed below the left eye and Fp1 (located above the left eye) was calculated. EEG analyses were done with MATLAB with functions from the EEGLAB (Delorme & Makeig, 2004) and ERPLAB (Lopez-Calderon & Luck, 2014) toolboxes, called by custom MATLAB code from our laboratory.

EEG data were high-pass filtered at 0.01 Hz and low-pass filtered at 30 Hz. EOG data were high-pass filtered at 0.1 Hz and low-pass filtered at 10 Hz. To remove blink-related ocular activity from the signal, an ICA (Makeig, Bell, Jung, & Sejnowski, 1996) was performed for each participant, and one or more components reflecting blink artifacts were removed from the continuous data (for details of the procedure used, see Drisdelle, Aubin, & Jolicoeur, 2017). An additional verification was performed on all epochs, after the ICA correction, to remove any remaining artifacts (vertical EOG deflection $> 50 \mu\text{V}$ within a time window of 150 msec) as well as all epochs containing horizontal eye movements (HEOG deflection $> 35 \mu\text{V}$ within a time window of 300 msec). Visual inspection of the averaged lateralized HEOG activity of each participant was then performed, and only participants with a deviation less than $5 \mu\text{V}$ toward the target (contralateral – ipsilateral activity) were kept for the final analysis. Moreover, for trials with seven or fewer channels containing artifacts (activity exceeding $\pm 100 \mu\text{V}$ in amplitude during the trial), those channels containing artifacts were interpolated, only for that trial, using the EEGLAB spherical spline interpolation function. When there were more than seven channels containing artifacts, the trial was excluded. Only trials with a correct response were included for the final analysis. Finally, epochs with RT outliers (determined using the Van Selst and Jolicoeur [1994] method) were also rejected. Three participants were not included in the final analysis because of technical difficulties during recording. For the remaining 151 participants, 55 participants were excluded because they had less than 50% of the total epochs (after epoch binning, and so half of the total trials in each average) remaining in at least one condition (near target – more frequent response, near target – less frequent response, far target – more frequent response, far target – less frequent response) or within-subjects RT tertiles (fast, middle, and slow trials) in either stimulus- or response-locked averages: 35 participants were removed because too many trials were lost during the automatic detection of horizontal eye movements; 18 participants, after inspection of the lateralized HEOG deflection (average saccade toward the target $> 5 \mu\text{V}$); and two, for too many trials rejected because of noise artifacts. Ninety-six participants were kept for the final analysis.

EEG Segmentation

Each trial produced two epochs, one segmented relative to stimulus onset and the other relative to the response. Epoch lengths for both stimulus- and response-locked averages were selected to include activity based on the group mean RT ($M = 657.4$ msec, $SD = 108.1$ msec): Stimulus-locked averaging used a time window from 200 msec prestimulus to 800 msec poststimulus, and response-locked averages used a time window from 1000 msec preresponse to 200 msec postresponse. Both stimulus- and response-locked segmentations were

baseline corrected using the mean voltage during the 200-msec period immediately before stimulus presentation.

Experimental Factors

Two analyses were conducted. In the first analysis, epochs were organized using a 2 × 2 repeated-measures experimental design consisting of the following factors: Frequency of response (more frequent or less frequent) and Distance of target from fixation (near or far). In the second analysis, EEG data were split into tertiles based on RT for each condition in the first analysis and then subsequently averaged to avoid any issues related to factor effects (factors: Frequency of response and Distance of target from fixation) for each participant on a trial-by-trial basis (within-subjects tertile split; i.e., trials were ranked by RT within each condition for each participant, split into tertiles, and aggregated by tertile) and also between participants, based on overall mean RT (between-subjects tertile split; i.e., participants ranked on overall mean RT).

Measurements and Statistics

The SPCN, N2pc, and RLpcN components are lateralized components calculated by subtracting ipsilateral activity (activity from electrodes over the left hemisphere for left targets and activity from electrodes over the right hemisphere for right targets) from contralateral activity (activity from electrodes over the left hemisphere for right targets and activity from electrodes over the right hemisphere for left targets). In the stimulus-locked average, the N2pc was quantified between 150 and 290 msec; and the SPCN, between 315 and 500 msec (see Figure 3). In the response-locked average, the RLpcN was observed from 700 msec before the response and up until the response was executed (0 msec; see Figure 4). Amplitude measurement windows were selected based on the individual peak of each condition or tertile (±30 msec centered at peak). Component latencies were calculated using a measure of fractional area latency on jackknife curves. Jackknife

latencies were back-transformed to the individual participant estimates (Smulders, 2010; Brisson & Jolicoeur, 2008; Kiesel, Miller, Jolicoeur, & Brisson, 2008). The onset of all components was calculated as the latency at which 20% of the area above the curve (and below 0 µV) was reached in their respective measurement window (from 150 to 290 msec for the N2pc, from 315 to 500 msec for the SPCN, and from −700 to 0 msec for the RLpcN). For all components, the maximum amplitude was observed at electrode pair PO7/PO8 (N2pc: −1.60 µV, SPCN: −1.75 µV, RLpcN: −1.37 µV), and so this electrode was used for analysis. The scalp distributions (collapsed across conditions) of the N2pc, the SPCN, and the RLpcN are depicted in Figure 2.

Statistical tests were performed with IBM SPSS (Version 24.0; SPSS Inc., 2016) and R (R Core Team, 2016) through the integrated development environment R-Studio (Version 1.0.136; R-Studio, 2016) using the following packages: ez (Lawrence, 2016), psych (Revelle, 2018), Hmisc (Harrell, 2018), tidyr (Wickham, 2017), and ppcor (Kim, 2015). Correlations and partial correlations (Pearson product-moment correlation coefficients) were calculated to evaluate the unique relationship between the N2pc and the RLpcN while controlling for the SPCN as well as between the SPCN and the RLpcN while controlling for the N2pc. The significance level of these correlations was adjusted for multiple comparisons using the false discovery rate method (Benjamini & Hochberg, 1995). ANOVA (with Greenhouse–Geisser correction when sphericity was violated in repeated-measures analyses) was used to evaluate the patterns of activity in stimulus- and response-locked epoch averages separately for each analysis and component as well as for behavioral analyses. Reported effect sizes for significant ANOVA results were determined using the generalized eta squared (η^2_G ; Bakeman, 2005; Olejnik & Algina, 2003). RT outliers were removed using the Van Selst and Jolicoeur (1994) method for each condition before splitting the EEG data based on RT tertiles. For the experimental manipulation analysis, Bonferroni-corrected pairwise comparisons were used

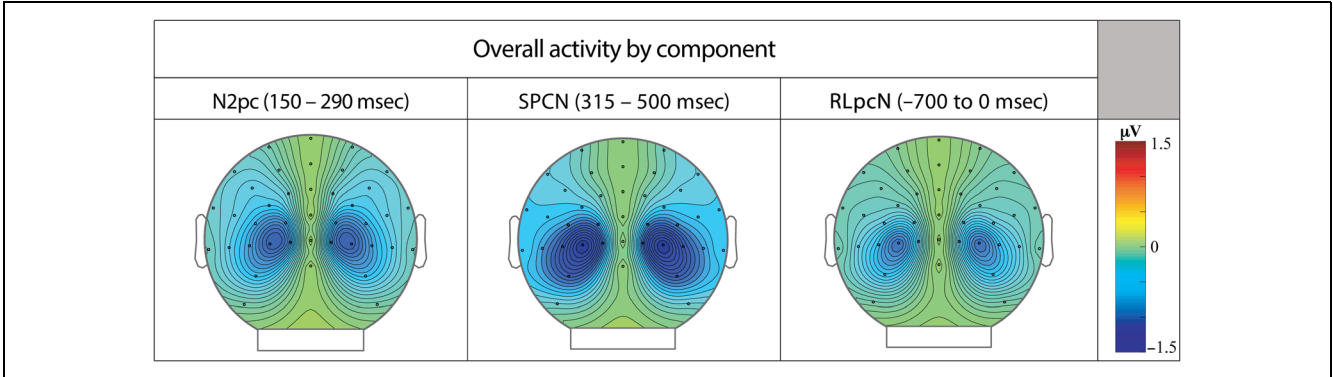


Figure 2. Lateralized (contralateral minus ipsilateral activity) grand-averaged rear view of scalp distributions for the N2pc (left; 150–290 msec after stimulus onset), SPCN (center; 315–500 msec after stimulus onset), and RLpcN (right; −700 to 0 msec before the response). Note that the left and right sides for all distributions are the same activity.

Table 1. Mean Accuracy (Mean Proportion Correct in the Target Discrimination Task; Acc.), Standard Deviation (*SD*), and Standard Error (*SE*) for Each Combination of Response Frequency (More Frequent or Less Frequent) and Distance from Fixation (Near or Far)

	More Frequent Response			Less Frequent Response		
	Acc.	<i>SD</i>	<i>SE</i>	Acc.	<i>SD</i>	<i>SE</i>
Near target	0.98	0.02	0.002	0.93	0.05	0.005
Far target	0.98	0.02	0.002	0.88	0.07	0.008

to decompose significant interactions when appropriate. When N2pc was split into tertiles by RT, differences between tertiles were assessed using two-tailed *t* tests with no corrections for multiple comparisons because we had an a priori hypothesis regarding the direction of the effect (an increase in amplitude or a decrease in latency with a decrease in RT; Loughnane et al., 2016; Clark et al., 2015; Wolber & Wascher, 2005). For the SPCN and RLpcN, Bonferroni-corrected pairwise comparisons were made, as in the decomposition of significant interactions in the experimental conditions analysis. Statistical analyses were conducted using a significance level of $p < .05$, with the exception of post hoc Bonferroni pairwise *t* tests comparisons, where a significance of $p < .025$ was used for all components in the experimental manipulation analysis (two comparisons), and a significance of $p < .017$ was used for the SPCN and RLpcN in the analysis where components were split into tertiles by RT (three comparisons).

RESULTS

Analysis 1: Comparison of Stimulus- and Response-locked Lateralized ERPs by Experimental Manipulations

Accuracy, RT, and ERP components were examined using a 2 (Response frequency: more frequent response, less frequent response) $\times$ 2 (Distance of target from fixation: near, far) repeated-measures ANOVA.

Behavioral Results

Mean accuracy was higher for more frequent than for less frequent responses ($F(1, 95) = 162.07, p < .0001, \eta_G^2 = .37$) as well as for targets near fixation than targets farther from fixation ($F(1, 95) = 77.57, p < .0001, \eta_G^2 = .07$; see Table 1). An interaction between Response frequency and Target distance from fixation conditions was observed ($F(1, 95) = 93.42, p < .0001, \eta_G^2 = .07$): Participants were more accurate for targets near fixation compared with targets farther from fixation when the task required the less frequent response ($t(95) = 9.59, p < .0001$), but not when it required the more frequent response

($t(95) = 0.45, p = .65$). Participants also responded faster when the task required the more frequent response compared with the less frequent response ($F(1, 95) = 85.13, p < .0001, \eta_G^2 = .04$) and for targets near fixation compared with targets farther from fixation ($F(1, 95) = 273.82, p < .0001, \eta_G^2 = .05$; see Table 2). An interaction between Response frequency and Distance from fixation conditions was observed ($F(1, 95) = 66.47, p < .0001, \eta_G^2 = .05$): RTs were shorter for targets near fixation compared with targets farther from fixation for both more frequent ($t(95) = 12.92, p < .0001$) and less frequent ($t(95) = 15.21, p < .0001$) responses, but there was a larger RT difference between targets presented near fixation and targets presented farther from fixation for the less frequent responses (68 msec) compared with the more frequent responses (35 msec).

Electrophysiological Results

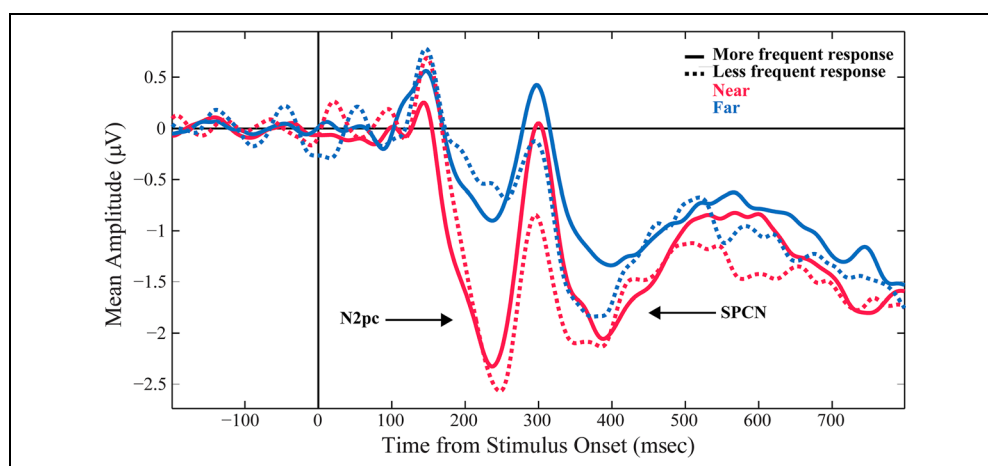
N2pc The N2pc separated by Frequency of response and Distance from fixation is shown in Figure 3. More frequent responses produced an earlier N2pc component by approximately 14 msec compared with less frequent responses ($F(1, 95) = 11.92, p = .0008, \eta_G^2 = .02$) but did not modulate N2pc amplitude ($F(1, 95) = 0.01, p = .91$). Targets nearer to fixation produced a significantly larger N2pc compared with targets farther from fixation ($F(1, 95) = 95.08, p < .0001, \eta_G^2 = .17$) but did not affect N2pc onset latency ($F(1, 95) = 0.82, p = .37$). An interaction between Frequency of response and Target distance from fixation was observed for N2pc amplitude ($F(1, 95) = 4.93, p = .03, \eta_G^2 = .003$), but not N2pc onset latency ($F(1, 95) = 0.03, p = .85$). The amplitude interaction was driven by a crossover effect: More frequent responses produced a slightly smaller N2pc than less frequent responses when the target was near fixation, whereas the opposite was found for targets farther from fixation (see Figure 3).

SPCN The SPCN waveforms for each combination of Frequency of response and Distance of target from fixation are shown in Figure 3. Less frequent responses produced a larger and earlier SPCN compared with more

Table 2. Mean RT (msec), Standard Deviation (*SD*), and Standard Error (*SE*) for Each Combination of Response Frequency (More Frequent or Less Frequent) and Distance from Fixation (Near or Far)

	More Frequent Response			Less Frequent Response		
	RT	<i>SD</i>	<i>SE</i>	RT	<i>SD</i>	<i>SE</i>
Near target	629	105	11	656	104	11
Far target	664	118	12	724	115	12

Figure 3. Lateralized (contralateral minus ipsilateral activity) grand-averaged waveforms time-locked to stimulus onset at electrodes PO7/PO8. The N2pc is observed from approximately 150 to 290 msec; and the SPCN, from approximately 315 to 500 msec poststimulus for each combination of response frequency (more frequent response [continuous traces] or less frequent response [dashed traces]) and distance from fixation (near fixation [red] or far from fixation [blue]). Amplitude measurement windows were selected based on the individual peak of each condition (± 30 msec).



frequent responses (amplitude: $F(1, 95) = 5.38, p = .02, \eta^2_G = .008$; latency: $F(1, 95) = 22.3, p < .0001, \eta^2_G = .04$). Targets near fixation produced a significantly larger SPCN compared with targets farther from fixation ($F(1, 95) = 10.22, p = .002, \eta^2_G = .02$). SPCN onset latency was not significantly different for near versus far targets ($F(1, 95) = 3.61, p = .06$). No interaction between Response frequency and Target distance from fixation was observed for SPCN amplitude ($F(1, 95) = 1.84, p = .18$) or onset latency ($F(1, 95) = 1.14, p = .29$).

RLpcN The lateralized waveforms time-locked to the response for each combination of Response frequency and Target distance from fixation can be seen in Figure 4. A larger RLpcN was found for trials with targets near fixation compared with trials with targets farther from fixation ($F(1, 95) = 7.86, p = .006, \eta^2_G = .01$), with a marginal difference in component onset latency ($F(1, 95) = 3.33, p = .07$). Less frequent responses produced a larger

RLpcN compared with more frequent responses ($F(1, 95) = 6.45, p = .01, \eta^2_G = .01$), and response frequency did not modulate onset latency ($F(1, 95) = 0.001, p = .97$). No interaction was observed between the Distance from fixation and the Response frequency manipulations for RLpcN amplitude ($F(1, 95) = 0.09, p = .76$) or latency ($F(1, 95) = 0.03, p = .85$).

Analysis 2: Comparison of Stimulus- and Response-locked Lateralized ERPs by Tertile Split on RT (Within-Subjects and Between-Subjects)

ERP components were examined using a 3 (Within-subjects tertile split by RT: fast trials, middle trials, slow trials) $\times$ 3 (Between-subjects tertile split by RT: fast participants, middle participants, slow participants) mixed-design ANOVA. Participant accuracy was analyzed using a one-way ANOVA (fast participants, middle participants, slow participants).

Figure 4. Lateralized (contralateral minus ipsilateral activity) grand-averaged waveforms time-locked to the response at electrodes PO7/PO8. The RLpcN is observed approximately 700 to 0 msec before the execution of a response for each combination of response frequency (more frequent response [continuous waveforms] or less frequent response [dashed waveforms]) and distance from fixation (near [red] or far [blue]). Amplitude measurement windows were selected based on the individual peak of each condition (± 30 msec).

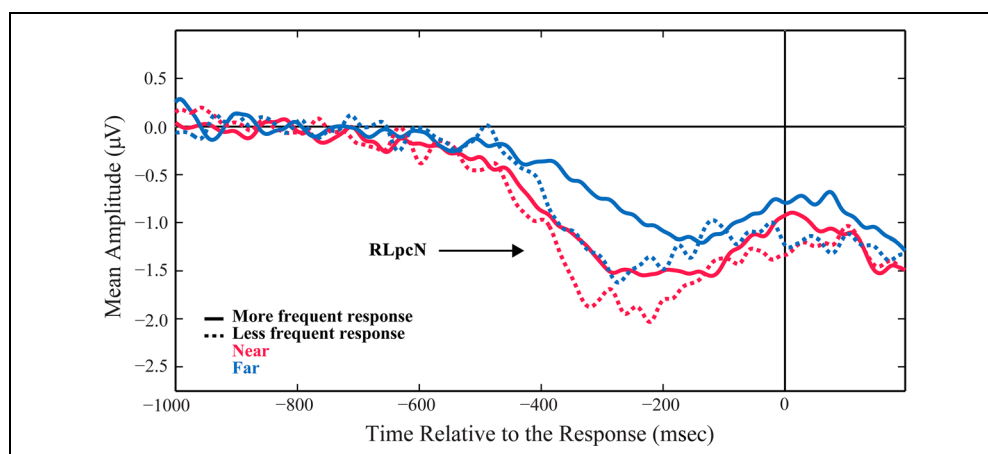


Table 3. Mean RT (msec) and Accuracy (Mean Proportion Correct in the Target Discrimination Task; Acc.) with Standard Deviation (SD) and Standard Error (SE) for Each Combination of Between-subjects (Fast Participants, Middle Participants, and Slow Participants) and Within-subjects (Fast Trials, Middle Trials, and Slow Trials) Tertiles

RTs	Fast Participants			Middle Participants			Slow Participants		
	RT	SD	SE	RT	SD	SE	RT	SD	SE
Fast trials (msec)	474	23	4	528	19	3	611	51	9
Middle trials (msec)	547	25	4	617	20	3	737	79	14
Slow trials (msec)	654	36	6	763	32	6	984	166	29
Accuracy	Acc.	SD	SE	Acc.	SD	SE	Acc.	SD	SE
Proportion correct	0.97	0.02	0.003	0.97	0.02	0.003	0.95	0.02	0.004

Behavioral Results

Slow participants were 2% less accurate than middle or fast participants ($F(2, 93) = 8.92, p < .0003, \eta^2_G = .16$; see Table 3). Mean RTs with standard errors and standard deviations split into tertiles on a trial-by-trial (within-subjects) and participant-by-participant (between-subjects) basis are given in Table 3.

Electrophysiological Results

N2pc The stimulus-locked contralateral minus ipsilateral difference wave showing the N2pc split into tertiles by RT for each combination of between-subjects and within-subjects tertiles is shown in Figure 5. When epochs were sorted into tertiles by trial RT (within-subjects), a difference in N2pc amplitude was observed ($F(2, 186) = 9.77, p = .0002, \eta^2_G = .02$), with no difference in N2pc onset latency ($F(2, 186) = 0.67, p = .51$). Planned comparisons demonstrated a significantly larger N2pc amplitude for shorter RTs between all three tertiles (fast-middle:

$t(95) = 2.23, p = .03$; fast-slow: $t(95) = 4.03, p = .0001$; middle-slow: $t(95) = 2.47, p = .02$; see Figure 6). When participants were sorted into tertiles by RT (between-subjects), a difference in N2pc latency was observed ($F(2, 93) = 3.78, p = .03, \eta^2_G = .06$), whereas the N2pc amplitudes were not significantly different ($F(2, 93) = 2.22, p = .11$). Planned comparisons showed an earlier N2pc onset with shorter overall mean RTs between fast and slow participants ($t(62) = 3.29, p = .002$), but not between fast and middle participants ($t(62) = 0.37, p = .71$) or middle and slow participants ($t(62) = 1.92, p = .06$). We did not find a Between Tertile $\times$ Within Tertile interaction either for the onset latency of the N2pc ($F(4, 186) = 0.81, p = .52$) or for the amplitude of N2pc ($F(4, 186) = 1.26, p = .29$).

SPCN The stimulus-locked contralateral minus ipsilateral difference wave showing the SPCN split by RT into tertiles for each combination of between-subjects and within-subjects tertiles is shown in Figure 5. A significant difference in SPCN onset latency was observed

Figure 5. Lateralized (contralateral minus ipsilateral activity) grand-averaged waveforms time-locked to stimulus onset at electrodes PO7/PO8. The N2pc is the first large negative deflection (with a peak between 150 and 290 msec), and the SPCN is the following large negativity (with peaks between 315 and 500 msec). Activity was separated into RT tertiles by participant average response speed (between-subjects; fast participants [solid curves], middle participants [larger dashed curves], and slow participants [smaller dashed curves]) and on a within-subjects, trial-by-trial basis (fast trials [blue curves], middle trials [green curves], and slow trials [red curves]). Amplitude measurement windows were selected based on the individual peak of each condition (± 30 msec).

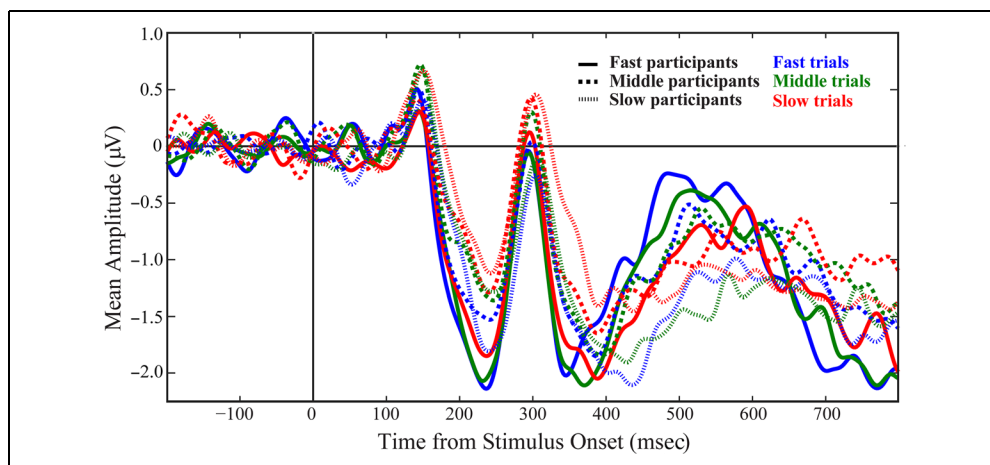
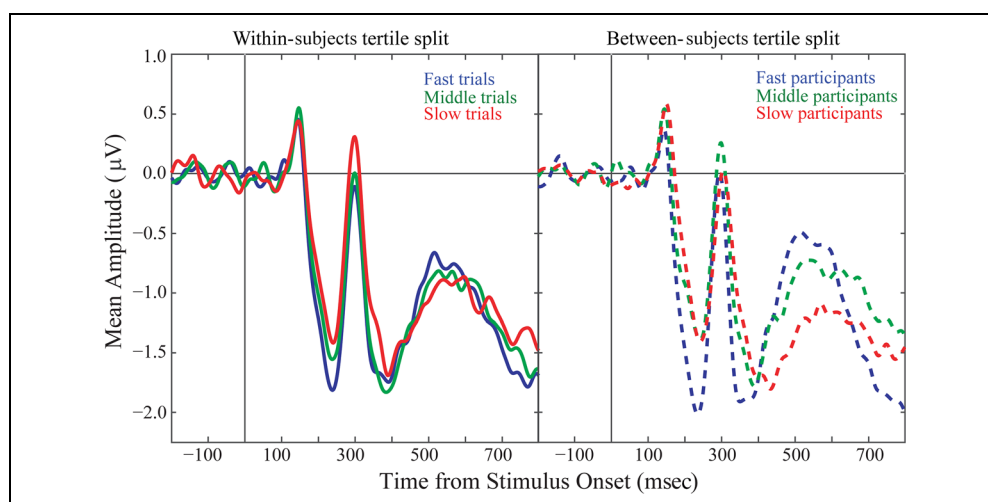


Figure 6. Lateralized (contralateral minus ipsilateral activity) grand-averaged waveforms time-locked to stimulus onset at electrodes PO7/PO8 showing the within-subjects (left, solid lines) and between-subjects (right, dashed lines) tertile split by RT effects separately for the stimulus-locked average. The N2pc is observed from approximately 150 to 290 msec poststimulus, and the SPCN, from approximately 315 to 500 msec. Activity is shown separately for fast (blue), middle (green), and slow (red) RT tertiles.

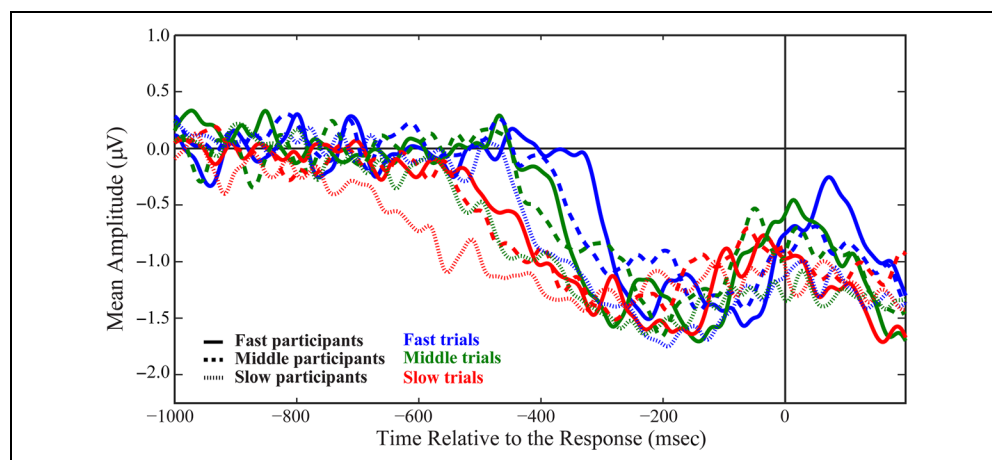


for both the subject-by-subject and trial-by-trial RT tertile splits (between: $F(2, 93) = 12.81, p < .0001, \eta_G^2 = .18$; within: $F(2, 186) = 22.01, p < .0001, \eta_G^2 = .04$; see Figure 6 for main effects). An interaction between Within-subjects and Between-subjects tertile splits by RT for SPCN onset latency was observed ($F(4, 186) = 2.63, p = .04, \eta_G^2 = .01$). The interaction was decomposed by performing three 1-way ANOVAs on the within-subjects (trial-by-trial) tertile split by RT separately for each between-subjects RT tertile. No significant difference in SPCN onset latency between fast, middle, and slow trial tertiles was observed for middle participants ($F(2, 62) = 1.23, p = .30$), whereas significant differences between trial RT tertiles were observed for both fast and slow participants (fast participants: $F(2, 62) = 17.93, p < .0001, \eta_G^2 = .11$; slow participants: $F(2, 62) = 11.20, p = .0004, \eta_G^2 = .06$). For both groups, paired t tests with corrections for multiple comparisons (Bonferroni-adjusted level of significance: $p < .016$) showed an earlier SPCN onset for fast trials compared with slow trials (fast participants: $t(31) = 5.93, p < .0001$; slow participants: $t(31) = 3.57, p = .001$) as well as for middle trials compared with slow trials (fast participants: $t(31) = 3.48, p = .002$; slow participants: $t(31) = 4.16, p = .0002$). No difference in SPCN latency between fast and middle trials was observed (fast participants: $t(31) = 2.50, p = .02$; slow participants: $t(31) = 1.06, p = .30$). No significant difference was observed in SPCN amplitude when data were sorted by RTs into tertiles for the between-subjects split ($F(2, 93) = 0.23, p = .80$). A significant difference in SPCN amplitude was observed, however, when trials (within-subjects) were sorted into RT tertiles ($F(2, 186) = 3.60, p = .03, \eta_G^2 = .006$). Paired t tests with corrections for multiple comparisons (Bonferroni-adjusted level of significance: $p < .016$) showed a significant difference between middle and slow trials ($t(95) = 2.94, p = .004$), but not between fast and middle trials ($t(95) = 0.50, p = .62$) or between fast and slow trials ($t(95) = 1.83, p = .07$).

An interaction between Within-subjects and Between-subjects RT tertiles was also observed for SPCN amplitude ($F(4, 186) = 2.67, p = .04, \eta_G^2 = .009$). The interaction was decomposed by performing three 1-way ANOVAs on the within-subjects (trial-by-trial) tertile split by RT separately for each of the between-subjects RT tertiles. No significant difference between fast, middle, and slow trials was observed for fast or middle participants (fast: $F(2, 62) = 1.31, p = .28$; middle: $F(2, 62) = 1.12, p = .33$). For slow participants, however, a significant difference in SPCN amplitude was observed across within-subjects tertiles split by RT ($F(2, 62) = 7.24, p = .002, \eta_G^2 = .04$). Paired t tests with corrections for multiple comparisons (Bonferroni-adjusted level of significance: $p < .016$) showed a larger amplitude for fast compared with slow trials ($t(31) = 3.62, p = .001$) as well as for middle compared with slow trials ($t(31) = 2.83, p = .008$), with no difference between fast and middle trials ($t(31) = 1.03, p = .31$).

RLpcN The RLpcN by RT tertile for each combination of within-subjects and between-subjects splits is shown in Figure 7, and the main effect waves are shown in Figure 8. As can be seen in these figures, the grand-averaged waveforms appeared to have clear latency differences and little or no amplitude differences. This was confirmed in the ANOVAs: RLpcN onset was earlier relative to the response (i.e., there was a longer period between component onset and the response) for longer RTs, $F(2, 186) = 75.1, p < .0001$, and $\eta_G^2 = .24$ for the within-subjects main effect and $F(2, 93) = 12.15, p < .0001$, and $\eta_G^2 = .14$ for the between-subjects main effect (Figure 8). For RLpcN amplitude across within-subjects RT tertiles, multiple comparisons (using the significance level of $p < .016$) showed a longer period between the onset of the RLpcN and the motor response for slower RTs across all levels (fast-middle: $t(95) = 4.9, p < .0001$; fast-slow: $t(95) = 11.81, p < .0001$;

Figure 7. Lateralized (contralateral minus ipsilateral activity) grand-averaged waveforms time-locked to the response at electrodes PO7/PO8. The RLpcN (response-locked contralateral negativity) reached a maximum negativity about 300 to 100 msec before the response. The waveforms show each combination of between-subjects RT tertiles (fast participants [continuous curves], middle participants [larger dashed curves], and slow participants [smaller dashed curves]) and within-subjects (trial-by-trial) RT tertiles (fast trials [blue], middle trials [green], and slow trials [red]). Amplitude measurement windows were selected based on the individual peak of each condition (± 30 msec).



middle–slow: $t(95) = 7.27, p < .0001$). For the RLpcN split into subject-by-subject RT tertiles, multiple comparisons ($p < .016$) showed an earlier RLpcN onset for slow compared with fast participants ($t(62) = 4.77, p < .0001$) and slow compared with middle participants ($t(62) = 3.00, p = .004$) with no difference between middle and fast participants ($t(62) = 1.68, p = .10$; Figure 8). There was no Between-subjects $\times$ Within-subjects interaction for latency ($F(4, 186) = 0.67, p = .61$). Finally, for the analysis of mean amplitudes, there were no significant effects ($F_s < 1$).

Difference between RT and RLpcN latency measures Although the onset latency effects for the RLpcN appear to represent a larger portion of the RT variance than for the N2pc or the SPCN, the latency differences between tertiles split

by RT only appear to account for approximately half of the behavioral RT differences (see Table 4). The apparent discrepancies among these measures are likely attributable to several factors. First, RT distributions were positively skewed, and so the mean was influenced by the positive tail of the distribution (i.e., slower trials), which could create a larger estimated difference between groups, in particular, the difference between middle and slow trials and participants. The smaller differences between tertiles split by RT when using median RTs as a measure of central tendency instead of mean RTs (see Table 4) suggest that this was a contributing factor. Moreover, the fractional-area latency component onset-latency measure used, that is, the latency at which 20% of the area from -700 to 0 msec before the response was accumulated (from left to right in the graphs), is a robust and

Figure 8. Lateralized (contralateral minus ipsilateral activity) grand-averaged waveforms time-locked to the response at electrodes PO7/PO8 showing the between-subjects (left, solid lines) and within-subjects (right, dashed lines) RT tertile split effects separately. The RLpcN is observed from approximately 700 to 0 msec before response execution. Activity is shown separately for fast (blue), middle (green), and slow (red) RT tertiles.

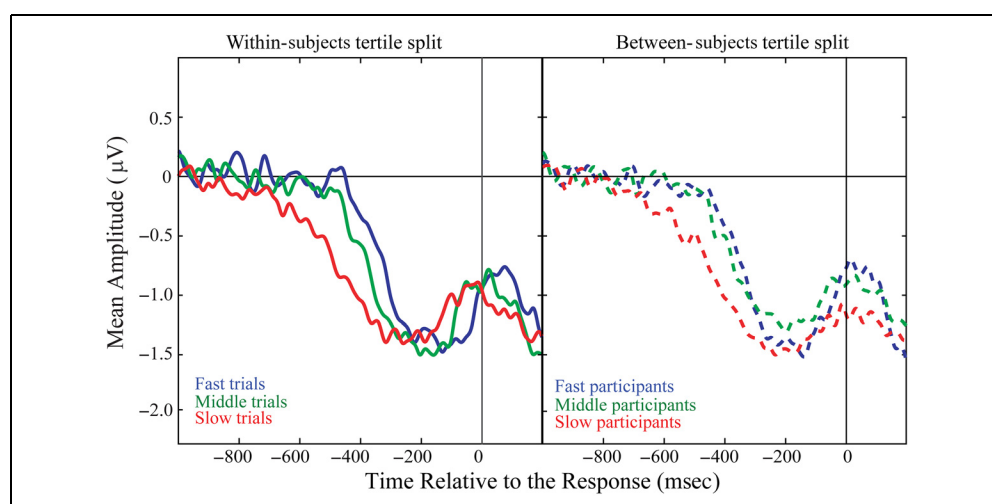


Table 4. N2pc, SPCN, and RLpcN Onset Latency Estimates (Fractional Area Latency [FAL] and Fixed Threshold Latency at $-0.75 \mu\text{V}$ [Fixed]) and RT (msec) Measures (Mean and Median) for Within-subjects and Between-subjects Tertile Splits by RT

	<i>Within-subjects Tertile Split</i>					
	<i>Mean RT</i>	<i>Median RT</i>	<i>N2pc (FAL)</i>	<i>SPCN (FAL)</i>	<i>RLpcN (FAL)</i>	<i>RLpcN (Fixed)</i>
Fast trials	538	528	202	357	−289	−350
Middle trials	634	618	204	360	−339	−405
Slow trials	801	757	205	367	−427	−497
Fast–middle	96	90	2	3	50	55
Middle–slow	167	139	1	7	88	92

	<i>Between-subjects Tertile Split</i>					
	<i>Mean RT</i>	<i>Median RT</i>	<i>N2pc (FAL)</i>	<i>SPCN (FAL)</i>	<i>RLpcN (FAL)</i>	<i>RLpcN (Fixed)</i>
Fast participants	559	548	198	349	−310	−360
Middle participants	636	618	200	360	−339	−399
Slow participants	777	713	214	374	−406	−492
Fast–middle	77	70	2	11	29	39
Middle–slow	141	95	14	14	67	93

Differences between fast and middle as well as middle and slow trials and participants are reported in the last two rows of each table section.

sensitive way to compare the onset of each curve relative to one another. Although this measurement algorithm possesses many advantages (Luck, 2014; Kiesel et al., 2008), for comparison purposes, we also measured the onset of the RLpcN by using the latency of each curve at a fixed amplitude value of $-0.75 \mu\text{V}$. This method replicated the patterns of ANOVA statistical results based on the fractional area latency reported above (within-subjects main effect: $F(2, 186) = 19.93, p < .0001, \eta^2_G = .11$; between-subjects main effect: $F(2, 93) = 11.14, p < .0001, \eta^2_G = .09$; interaction: $F < 1$) but yielded slightly larger onset differences in RLpcN latency between tertiles (see Table 4).

Correlations between overall stimulus-locked activity and response-locked activity Correlations were calculated to understand better the extent to which the RLpcN reflected activity leading to the N2pc and SPCN components. The relationship between stimulus- and response-locked averages was assessed by calculating partial correlations between the N2pc and the RLpcN while controlling for SPCN and between the SPCN and the RLpcN while controlling for N2pc for each within-subjects tertile split by RT (fast trials, middle trials, slow trials) separately (see Table 5). We correlated the mean activity in 60-msec time windows for the N2pc (205–265 msec for the fast and middle trial tertiles and 210–270 msec for the slow trial tertile) and SPCN (360–420 msec for the fast and slow trial tertiles and 350–410 msec for the middle trial tertile) with different 60-msec time windows

from the beginning to the end of the RLpcN (in steps of 60 msec, i.e., from -360 msec for fast trials, from -480 msec for middle trials, and from -660 msec for slow trials, until the response) to assess which portions of the RLpcN associated with the N2pc and the SPCN.

The patterns of correlations when looking at fast, middle, and slow trials separately suggest that the variance in RLpcN onset is an important factor to consider when examining the relationship between the stimulus- and response-locked activity (see Table 5). A larger partial correlation for N2pc compared with SPCN was observed near the onset of the RLpcN for fast and middle trials, demonstrating that the N2pc explains more unique variance than the SPCN in the earlier portion of the RLpcN. For all tertiles, the SPCN explained more unique variance than the N2pc in the later portion of the RLpcN.³ Slow trials, which are more variable in RT (see Table 3), show no significant partial correlations for N2pc, possibly suggesting that it does not contribute to the variance independently of the SPCN variance (see Table 5). However, given that, for fast and middle trials, we do observe a portion of the RLpcN curve associated with the N2pc and the slow trials demonstrate a stimulus-locked N2pc (see Figure 6), the lack of partial correlations was likely because of a large amount of variance between trials, jittering out the N2pc activity contributing to the RLpcN. Taken together, these results suggest that the RLpcN is a combination of both the N2pc and the SPCN and that they are more or less identifiable within the curve depending on the variability of RT.

Table 5. Correlations and Partial Correlations for N2pc and SPCN Components Separated into Fast, Middle, and Slow Trials

	<i>RIpcN Measurement Window (msec)</i>											
	-630 ± 30	-570 ± 30	-510 ± 30	-450 ± 30	-390 ± 30	-330 ± 30	-270 ± 30	-210 ± 30	-150 ± 30	-90 ± 30	-30 ± 30	
<i>Fast trials</i>												
N2pc (<i>r</i>)						.57*	.73*	.71*	.62*	.54*	.43*	
SPCN (<i>r</i>)						.49*	.74*	.73*	.86*	.82*	.73*	
N2pc (<i>pr</i> SPCN) ^a						.39*	.53*	.50*	.26*	.10	-.02	
SPCN (<i>pr</i> N2pc) ^a						.22*	.55*	.54*	.77*	.74*	.66*	
<i>Middle trials</i>												
N2pc (<i>r</i>)				.24*	.47*	.64*	.62*	.38*	.40*	.30*	.13	
SPCN (<i>r</i>)				.13	.36*	.65*	.73*	.78*	.78*	.70*	.52*	
N2pc (<i>pr</i> SPCN)				.20	.35*	.43*	.35*	-.16	-.1	-.18	-.26*	
SPCN (<i>pr</i> N2pc)				-.01	.12	.45*	.58*	.75*	.74*	.67*	.56*	
<i>Slow trials</i>												
N2pc (<i>r</i>)	.25*	.25*	.33*	.43*	.49*	.55*	.48*	.48*	.40*	.26*	.21*	
SPCN (<i>r</i>)	.32*	.36*	.45*	.63*	.75*	.80*	.79*	.79*	.70*	.55*	.49*	
N2pc (<i>pr</i> SPCN)	.05	.03	.06	.04	-.01	.07	-.06	-.06	-.1	-.15	-.17	
SPCN (<i>pr</i> N2pc)	.22	.26*	.32*	.51*	.66*	.69*	.71*	.71*	.63*	.52*	.48*	

Partial correlations are of the N2pc with the RIpcN controlling for the SPCN and of the SPCN with the RIpcN controlling for the N2pc. Calculations are based on 60-msec time windows at the peak of the stimulus-locked components (N2pc: fast [205–265 msec], middle [205–265 msec], and slow [210–270 msec]; SPCN: fast [360–420 msec], middle [350–410 msec], and slow [360–420 msec]) and 60-msec time windows of the RIpcN component. Because RIpcN onset latency was highly sensitive to RT variability, there are a variable number of correlations for each tertile (e.g., the component has a longer duration in slower trials; see Figure 8).

^a*pr* SPCN means partial correlation controlling for SPCN; *pr* N2pc means partial correlation controlling for N2pc.

*Significant at $p < .05$ using the Benjamini–Hochberg false discovery rate method to control for Type 1 errors because of multiple comparisons.

The **bold** numbers in the table represent significant correlations.

DISCUSSION

The goal of this study was to examine how stimulus- and response-locked lateralized averages of posterior attentional ERPs, namely, the N2pc, SPCN, and RLpcN components, can be used to understand sources of RT variability within the flow of information processing from stimulus to response. We compared the conventional stimulus-locked components (N2pc and SPCN) with response-locked (RLpcN) averages using electrodes where activity was maximal, which was the same for all components (PO7/PO8). The stimulus-locked N2pc component was observed from approximately 150 to 290 msec, followed by the stimulus-locked SPCN component from approximately 315 to 500 msec. Response-locked averages consistently produced a posterior contralateral negative deflection, which we call the response-locked contralateral negativity (RLpcN), characterized by an onset from approximately 700 msec before the response, followed by a partial offset (not completely returning to baseline) at the moment of response. There were significant partial correlations for the N2pc and SPCN components with the RLpcN, where the N2pc had a stronger relationship with the early portion of the component (compared with the SPCN) and the SPCN had a stronger relationship with the peak amplitude (compared with the N2pc), closer to the response. Moreover, the scalp distribution of the overall RLpcN was very similar to those of the N2pc and SPCN (see Figure 2).

The first set of analyses examined data sorted by experimental manipulations: The target was either nearer or farther from fixation, and the response was either more frequent (75% of target-present trials) or less frequent (25% of target-present trials). The second set of analyses sorted EEG data into tertiles by RT (fast, middle, or slow) on a within-subjects (trial-by-trial) basis and on a between-subjects basis. We hypothesized for both types of analyses that the engagement of spatial attention would be indexed by the onset of a posterior contralateral negativity (N2pc for stimulus-locked segmentations or RLpcN for response-locked segmentations). Consequently, whereas the onset of the N2pc reflected the time at which perceptual attention engaged on the target, the time between the onset of the RLpcN and the response reflected the duration of processing during and after the engagement of attention. Results from the tertile split on RT analyses showed that the time between RLpcN onset and when the response was made increased with increasing RT. Furthermore, our hypothesis is supported by the relatively small latency effects in N2pc and SPCN onsets when data were split into tertiles by RT, which suggests that a large portion of processing determining RT occurred after the onset of spatial attention. Importantly, the RLpcN was sustained and correlated with the SPCN amplitude until the response, and their amplitudes behaved in the same way in almost all cases. Our results therefore suggest that the subsequent pro-

cessing of visual information or representations are maintained up until (and partially after) response execution. In particular, results from both analyses (separated by experimental condition or into tertiles sorted by RT) also showed that RLpcN amplitude was larger when the task was more likely to require the sustained maintenance of visual information, similar to the stimulus-locked SPCN. On the basis of our results, we conclude that, in the present work, the RLpcN reflects the N2pc and SPCN from a different perspective, that is, from the onset of the N2pc to the response.

Analysis 1: Experimental Manipulations

The visual search task consisted of a 2 (target near fixation vs. target far from fixation) $\times$ 2 (target required a more frequent or less frequent response) experimental design.

As found in previous work, N2pc amplitude was larger when targets were closer to fixation (see Figure 3; West et al., 2015; Schaffer et al., 2011). The same pattern (an increase in amplitude with a decrease in target eccentricity) was observed for the SPCN and RLpcN, suggesting a similar neural support to that of the N2pc and that the underlying mechanisms are not strongly time-jittered relative to stimulus presentation (because it was observed in both averages), but rather a more general increase in activity when visual items are closer to fixation.

Frequency of response modulated the N2pc and SPCN components' onset latency: Although more frequent response trials produced an earlier N2pc, they produced a delayed SPCN. For the N2pc, this result appears at first puzzling because the frequency of the response was expected to affect processing after the engagement of attention (i.e., you would have to attend to and select the object of interest before deciding on a response). One possibility was that this difference in onset latency of the N2pc may be a by-product of averaging differing numbers of trials (there are three times more trials in the more frequent response condition than in the less frequent response condition) and that averages based on more trials are more likely to include more fast N2pc onsets (based on the statistics of extremes; see Logan, 1988). However, we ruled out this possibility by computing three separate averages for the more frequent response condition, each based on one third of trials. The mean onset latency difference for the N2pc between the more frequent response averages and the less frequent response average was about 12 msec (down from 14 msec when all of the more frequent response trials were included in the same average). Hence, most of the latency difference could not be accounted for by appeal to averaging different numbers of trials. We thus speculate that engagement on the target was influenced by both color and shape cues, rather than only color, as originally predicted, although we believe that the most important target selection cue was color. We speculate

that a gap at the top of a square (associated with the less frequent response) delayed full attentional engagement, whereas a gap at one of the other sides (associated with the more frequent response) accelerated attentional engagement based on the statistical association of gap location with the response. Indeed, the difference in the latency of N2pc between both conditions (more frequent and less frequent responses) accounted for approximately 30% of RT ($12\text{--}14\text{ msec} \div 44\text{ msec}$). For the SPCN, the opposite result was found: Less frequent response trials produced an earlier component onset compared with more frequent response trials. We suggest that this earlier SPCN onset could reflect an earlier need for visual representation processing in VSTM for less frequent response trials. Previous research has suggested that the SPCN is modulated by a need for support from VSTM for downstream processing of visual inputs, and perhaps displays leading to a less frequent response engaged VSTM sooner (see Prime & Jolicoeur, 2010).

Finally, trials requiring a less frequent response produced a significantly larger RLpcN and SPCN in comparison with trials requiring a more frequent response. We suggest that activity associated with the less frequent response was larger because executing this response was a less prepared process, and a more neural activity, possibly reflecting an increase in the need for subsequent processing after object selection in VSTM, was recruited. We had, however, also expected that more time would pass between the onset of the RLpcN and the response for trials requiring a less frequent response, demonstrating an increase in processing time required following the deployment of attention. However, no significant difference in RLpcN latency was observed between both frequency of response conditions. Given the small (albeit significant) difference in the behavioral RT measure (approximately 44 msec) between more frequent and less frequent response trials, it is possible that, although a more detailed representation was maintained for less frequent response trials, producing a larger RLpcN and SPCN, the task was not sufficiently difficult, or these trials sufficiently rare, to observe a difference in RLpcN onset. Importantly, this effect does not appear to be because of differences in RT variance between both conditions, which would affect onset (more or less variable in time) and thus the averaging of trials, as the standard deviations and standard errors for behavioral RTs are nearly identical for both response frequency options (see Table 2). More research would be required on the subject to understand the association between stimulus- and response-locked averaging for this experimental manipulation.

Analysis 2: ERPs by Tertile Split on RT (Within-Subjects and Between-Subjects)

The second analysis sorted epochs into tertiles by RT: Data were sorted within-subjects, based on individual

trial RT, and between-subjects, based on the overall mean RT of each participant.

A significantly earlier N2pc for fast participants compared with slow participants was observed, with no difference in component onset across within-subjects RT tertiles, suggesting little or no trial-to-trial variation, but a difference between faster and slower individuals. Moreover, N2pc amplitude increased with decreasing within-subjects tertile response speeds (Figures 5 and 6), as observed in a previous study using a large portion of the same data (Drisdelle et al., 2016), suggesting that differences in electrophysiological activity associated with the eventual RT occurred as early as the N2pc. This, in turn, possibly suggests that a more efficient deployment of attention (indexed by a larger N2pc amplitude) sped up subsequent processing. It is also possible, however, that RT and N2pc were modulated by a third factor and that N2pc is not on the neural pathway responsible for variations in RT. Although we cannot rule out alternative accounts, shorter RTs were also associated with an earlier SPCN component in the present work, suggesting that visual information began subsequent processing in VSTM earlier, and previous research also showed correlations between RT with N2pc amplitude (Loughnane et al., 2016) and N2pc latency (Wolber & Wascher, 2005). An explanation in terms of more efficient processing, as suggested by the increase in N2pc amplitude, is therefore at least consistent with the results. More research will be needed to test this hypothesis.

In a previous report with the same task and many of the same participants (Drisdelle et al., 2016), we found significant between-subjects and within-subjects amplitude effects for the N2pc when EEG data were split based on median RT (i.e., shorter RTs produced a larger N2pc compared with longer RTs), whereas the present work only replicated the within-subjects effect just described. One possible explanation is that the between-subjects effect was not as robust as the within-subjects effect because of interindividual variance. Boy and Sumner (2014) suggested that between-subjects variability can arise from an entirely different source than that which is driving the within-subjects effects (see also Borsboom, Kievit, Cervone, & Hood, 2009). Across several visual priming tasks, the authors observed a strong and systematic relationship for their within-subjects priming manipulations and no between-subjects effects. They concluded that this is because between-subjects variance depends on both study parameters and participant idiosyncrasies. Although the waveforms in Figure 6 suggest a similar pattern for both splits (larger N2pc for faster trials and participants), the between-subjects test fell short of significance, probably because of lower statistical power to detect an effect in comparison with the within-subjects test.

When considering both the within-subjects and between-subjects RT tertile analyses, we found a significant difference in SPCN amplitude by tertile split on

RT between fast and slow trials as well as middle and slow trials, but only for slow participants, where SPCN was smaller for slower trials. We speculate that this difference is because of a special circumstance for the slow participants. We tentatively suggest that the slow trials for slow participants were particularly slow because of a difficulty in extracting relevant information during object selection (producing a smaller N2pc), which in turn lead to a smaller SPCN due to less information available for any necessary subsequent processing. This, however, does not appear to be the case for the other levels of the tertile splits by RT. This result could also suggest that, although the amplitudes of the SPCN and RLpcN are reflecting the same underlying neural mechanisms involved in visual information processing, the amplitude of SPCN may be a more sensitive measure of post-N2pc processing difficulty. Finally, SPCN onset latency was generally earlier for faster RTs compared with slower RTs, which we suggest is because of a speeding up of processing, leading to an earlier passage of objects into VSTM (see Figure 6). It is important to note, however, that these SPCN onset differences were small (see Table 4).

Because of the large differences in RT between tertiles (see Table 4) and the corresponding small (between-subjects) or not significant (within-subjects) shifts in N2pc onset, the results suggest that most of the processing speed variations in the present paradigm occurred downstream of the initial deployment of visual spatial attention, in line with previous work (Cosman et al., 2016; Drisdelle et al., 2016; Clark et al., 2015; Töllner et al., 2012; Hackley et al., 2007; Wolber & Wascher, 2005). In addition, the relatively small, yet significant, differences observed in SPCN onset also cannot explain the large RT differences observed between tertiles, suggesting that most of the processing accounting for variations in RT occurred after the onset of subsequent target processing in VSTM as well. For this reason, we would expect a large portion of the RT difference to be reflected in RLpcN onset latency (i.e., processing after the initial deployment of attention). This is what was observed: A much larger latency effect was observed for the onset of the response-locked component (RLpcN; see Figures 7 and 8) in comparison with the onset of the stimulus-locked components (N2pc and SPCN; see Figures 5 and 6). A comparison of behavioral RT medians instead of means (attenuating the effect of positively skewed RT distributions) and jackknifed latencies measured at $-0.75 \mu\text{V}$ showed that much of the difference in RT between tertiles can be accounted for by the corresponding latencies differences in the RLpcN (see Table 4).

Both SPCN and RLpcN amplitudes did not generally differ across RT tertiles (with some exceptions in the SPCN, described above) and were modulated in the same way for the experimental manipulations, corroborating the idea that the amplitude of both components represents the same activity from different perspectives. This claim is further corroborated by the strong partial

correlations (unique variance when the N2pc variance was controlled for) between the amplitude of the later portion of the RLpcN and the amplitude of the SPCN (see Table 5). In other words, the amplitudes of these components appear to both reflect the representation of information in VSTM, whereas the leading edge of the RLpcN is reflecting the initial deployment of attention (N2pc onset). A prediction of this hypothesis is that the amplitude of the RLpcN should be affected by the number of items stored in VSTM, similar to the SPCN (Vogel & Machizawa, 2004). We could not verify this with the present experiment but plan to test in future work. The results from the tertile-split-by-RT analysis, on the other hand, with large delays between RLpcN onset and the response and very small differences in N2pc and SPCN onset latencies, suggest that much of the RT variance reflects processes that follow the onset of visual spatial attention and the subsequent processing in VSTM, highlighting the usefulness of the present methods.

To summarize, the three most important findings regarding the relationship between ERP and RT latencies in the present work were the following. First, the N2pc had a larger amplitude for shorter RT trials and an earlier onset for faster participants, demonstrating that differences in electrophysiological activity associated with eventual RTs occur as early as the engagement of attention. Second, SPCN latency was also slightly earlier for shorter RTs, suggesting that after attentional deployment, subsequent processing of visual information was sped up. Third, although small latency effects were observed for both the N2pc and the SPCN, the tertile split analysis demonstrated much larger differences in latency across RT tertiles for the RLpcN, suggesting that the resulting RT was likely in majority modulated by cognitive processes taking place after the initial engagement of attention, likely while task-relevant representations were held in VSTM.

Conclusion

The RLpcN onset results of the tertile-split-by-RT analysis strongly suggest that the moment of an eventual response was determined by processing occurring after the deployment of attention, in line with previous research (Cosman et al., 2016; Töllner et al., 2012; Hackley et al., 2007). However, that is not to say that the N2pc and SPCN do not potentially play a role: Faster trials were associated with a larger N2pc and an earlier SPCN. Moreover, in nearly all analyses, the amplitude of the RLpcN was modulated similarly to the SPCN, suggesting that for the present task, visual representation in VSTM did not fade during later stages of processing but was maintained (RLpcN was sustained until the response). Future research should consider a task with longer and more variable RTs to confirm the involvement of the RLpcN in VSTM, and possibly response selection, and whether this component is exclusively a representation

of SPCN and N2pc or contains other activities not modulated in the present task. Nevertheless, the RLpcN appears to reflect both the initiation (N2pc) and subsequent processing of selected objects in VSTM (SPCN) as well as the duration of processing after the deployment of visual spatial attention. The combined examination of N2pc, SPCN, and RLpcN provides a powerful method to determine the locus of RT variability within the stream of information processing, relative to the onset of visual-spatial attention.

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Notes

1. In the work of Töllner et al. (2012), a compound response (or task) was defined as requiring a motor response that was determined independently of the target defining dimensions (e.g., a target is defined by either color or shape and a response based on target orientation).
2. This component has also been referred to as contralateral delay activity (CDA; Vogel & Machizawa, 2004) and the contralateral negative slow wave (CNSW; Klaver, Talsma, Wijers, Heinze, & Mulder, 1999).
3. For middle trials, from 60 msec to the response, a negative and significant partial correlation was observed between the N2pc and the RLpcN ($r = -.26, p = .02$), whereas this relationship was positive, but not significant, in the original correlations ($r = .13, p = .22$). Given the redundancy between N2pc and SPCN ($r = .58, p < .0001$) and the small numerical increase in correlation (controlling for N2pc) between the SPCN and this time window of the RLpcN (from .52 for the correlation to .56 for the partial correlation), the N2pc may be acting as a suppressor variable. We put forward a possible scenario: When the variance associated with the SPCN is controlled for, processes terminate earlier (returning to baseline, marked by a decrease in contralateral activity) following a more efficient allocation of attention (an increase in N2pc amplitude). However, given that the main focus of this analysis was to demonstrate that the RLpcN reflects a combination of both the N2pc and the SPCN and that this was only significant for middle trials, we do not further elaborate on this result.

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